STUDY GUIDE

Thyroid Physiology & Disorders

Anatomy · Hormone Synthesis · Regulation · Clinical Disorders

▸ Anatomy of the Thyroid Gland
▸ Thyroid Hormone Production
▸ Thyroid Hormones: Chemistry & Transport
▸ Physiological Effects of Thyroid Hormones
▸ Regulation of Thyroid Hormone Secretion
▸ Mechanism of Action
▸ Disorders of the Thyroid Gland

everythingabuad.com

PHS 220 — Physiology

§ TABLE OF CONTENTS

1. Anatomy of the Thyroid Gland
    · Gross anatomy
    · Microscopic structure
    · Blood supply
2. Thyroid Hormone Production
    · Iodide trapping
    · Organification
    · Hormone storage & release
3. Thyroid Hormones: Chemistry & Transport
    · T3 vs T4
    · Transport proteins
    · Peripheral conversion
4. Physiological Effects of Thyroid Hormones
    · Metabolic effects
    · Cardiovascular
    · CNS & development
    · Other systems
5. Regulation of Thyroid Hormone Secretion
    · HPT axis
    · Autoregulation by iodine
    · Exogenous factors
6. Mechanism of Action
    · Nuclear receptor signalling
    · Non-genomic effects
7. Disorders of the Thyroid Gland
    · Hyperthyroidism
    · Hypothyroidism
    · Goitre
    · Thyroid nodules

§1 ANATOMY OF THE THYROID GLAND

The thyroid gland is a highly vascularised, butterfly-shaped endocrine organ situated in the anterior neck. It is one of the largest endocrine glands in the body, weighing approximately 20–30 g in adults. Critically, its normal function is essential for every cell in the body, as thyroid hormones are required for basal metabolic activity.

1.1 Gross Anatomy

Feature	Description
Shape	Butterfly-shaped (two lateral lobes joined by isthmus); a pyramidal lobe is present in ~50% of individuals
Position	Anterior to the 2nd–4th tracheal rings; deep to the sternothyroid and sternohyoid muscles
Relations	Posterolateral — carotid sheath; Posterior — parathyroid glands (×4), recurrent laryngeal nerves
Weight	20–30 g (adults); increases during pregnancy and iodine deficiency
Arterial supply	Superior thyroid artery (from ECA) + Inferior thyroid artery (from thyrocervical trunk)
Venous drainage	Superior, middle, inferior thyroid veins → IJV / brachiocephalic vein
Lymphatics	Pretracheal, paratracheal, and deep cervical nodes
Nerve supply	Vasomotor supply from superior cervical ganglion; no secretomotor innervation (TSH-driven)

1.2 Microscopic Structure

The functional unit of the thyroid is the **follicle (acinus)** — a spherical structure lined by a single layer of follicular epithelial cells (thyrocytes) surrounding a lumen filled with **colloid** (predominantly thyroglobulin). The thyroid contains approximately 3 million follicles.

Cell Type	Location	Function
Follicular cells (Thyrocytes)	Follicle wall	Synthesise and secrete T3 and T4; trap iodide; produce thyroglobulin
Colloid	Follicle lumen	Storage depot for thyroglobulin-bound thyroid hormones (2–3 months supply)
Parafollicular cells (C-cells)	Interfollicular stroma	Synthesise and secrete Calcitonin in response to hypercalcaemia

HISTOLOGY NOTE

> ***The state of the follicle reflects its activity level. An inactive follicle has large colloid and flat (squamous) thyrocytes. An active follicle has small colloid and tall cuboidal-to-columnar thyrocytes — the increased height reflects upregulation of secretory machinery in response to TSH.***

§2 THYROID HORMONE PRODUCTION

Thyroid hormone synthesis is a unique and complex multi-step process that requires iodine — an essential trace element supplied entirely from the diet. The daily iodine requirement for adequate thyroid function is approximately 150 µg in adults (200 µg in pregnancy). The entire process occurs at the interface of the thyrocyte apical membrane and the colloid.

2.1 Step-by-Step Synthesis

Step 1: Iodide Trapping (Active Uptake)

Iodide (I^-) is actively transported from the blood into the thyrocyte against an electrochemical gradient by the Na^+/I^- Symporter (NIS) located on the basolateral membrane. NIS co-transports 2 Na^+ ions per I^- ion, using the Na^+ gradient maintained by Na^+/K^+-ATPase as the driving force. This concentrates iodide 30–250× above plasma levels. NIS is the target of radioiodine (^{131}I) therapy in thyroid cancer and hyperthyroidism.

Step 2: Iodide Efflux into Colloid

Iodide is transported across the apical membrane into the colloid by Pendrin (SLC26A4), a Cl^-/I^- antiporter. Mutations in the Pendrin gene cause Pendred syndrome (congenital hypothyroidism + sensorineural deafness).

Step 3: Thyroglobulin Synthesis

Thyroglobulin (Tg) — a large glycoprotein (MW ~660 kDa, dimer) — is synthesised by the rough ER, glycosylated in the Golgi, and exocytosed into the colloid. Tg contains ~140 tyrosine residues, of which ~25–30 are iodinated. It is the scaffold on which thyroid hormones are synthesised and stored.

Step 4: Oxidation of Iodide → Active Iodine

Iodide (I^-) is oxidised to reactive iodine (I^0 or I^+) by Thyroid Peroxidase (TPO) at the apical membrane, in the presence of H_2O_2 (generated by Dual oxidase, DUOX2). This step is inhibited by propylthiouracil (PTU) and carbimazole/methimazole.

Step 5: Organification (Iodination of Thyroglobulin)

Active iodine is rapidly incorporated onto the phenol rings of tyrosine residues within Tg, also catalysed by TPO. This produces Monoiodotyrosine (MIT) and Diiodotyrosine (DIT). Organification is also blocked by PTU and carbimazole.

Step 6: Coupling (Formation of T3 and T4)

Two iodotyrosines on Tg are coupled (by TPO) to form the iodothyronines: • MIT + DIT = Triiodothyronine (T3) — 3 iodine atoms • DIT + DIT = Thyroxine (T4) — 4 iodine atoms This coupling reaction forms T3 and T4 while they remain covalently bound within the Tg molecule in the colloid.

Step 7: Storage

Thyroid hormones remain stored within the colloid, still attached to Tg, for up to 2–3 months. This large reserve protects against short-term iodine deficiency and explains why antithyroid drugs take weeks to normalise hormone levels.

Step 8: Secretion

Upon TSH stimulation, thyrocytes reabsorb colloid by pinocytosis. Lysosomes fuse with the resulting endosomes and protease enzymes (cathepsins) cleave T3 and T4 from Tg. Free T3 and T4 are then secreted across the basolateral membrane into the bloodstream. Residual MIT and DIT are deiodinated by iodotyrosine dehalogenase, recycling iodide within the cell.

> **DRUG TARGET SUMMARY**
>
> **PTU (Propylthiouracil) & Carbimazole/Methimazole:** Block TPO → inhibit organification and coupling. PTU also inhibits peripheral T4→T3 conversion (preferred in thyroid storm and 1st trimester pregnancy).
> **Radioiodine (^{131}I):** Taken up by NIS; β-radiation destroys thyroid follicular cells. Used for Graves' disease and differentiated thyroid cancer.
> **Lithium:** Blocks Tg proteolysis and hormone release; used rarely in severe thyrotoxicosis.
> **Iodine excess (Wolff-Chaikoff effect):** Acute iodine loading transiently inhibits thyroid hormone synthesis — protective mechanism.

§3 THYROID HORMONES: CHEMISTRY & TRANSPORT

3.1 T3 vs T4 — Key Comparisons

Property	T4 (Thyroxine)	T3 (Triiodothyronine)
Iodine atoms	4	3
Secretion ratio (from thyroid)	~80%	~20%
Potency	1× (reference)	3–4× more potent than T4
Plasma half-life	~7 days	~1 day
Plasma concentration	~100 nmol/L (higher)	~2 nmol/L (lower)
Protein binding (TBG)	~99.97% bound	~99.7% bound
Free fraction (active)	~0.03%	~0.3%
Role	Main circulating form; prohormone for T3	Principal active form at receptor level

Property	T4 (Thyroxine)	T3 (Triiodothyronine)
Peripheral conversion	Converted to T3 by 5'-deiodinase	Active form; also produced as rT3 (inactive)

3.2 Plasma Transport Proteins

Protein	Abbreviation	% T4 Carried	% T3 Carried	Notes
Thyroxine-Binding Globulin	TBG	~70%	~80%	Main carrier; highest affinity; levels ↑ with oestrogen (pregnancy, OCP) ↓ with androgens
Transthyretin (Prealbumin)	TTR	~15%	~5%	Also carries retinol-binding protein
Albumin	HSA	~15%	~15%	Low affinity but high capacity; most abundant plasma protein

CLINICAL PEARL — Total vs Free Thyroid Hormones

Changes in TBG levels alter **total** thyroid hormone concentrations but not **free** (active) fractions or clinical thyroid status. Pregnancy and oestrogen therapy raise TBG → higher total T4, but free T4 remains normal and the patient is euthyroid. Always interpret total T3/T4 in the context of TBG levels; or measure free T4 (FT4) and free T3 (FT3) directly.

§4 PHYSIOLOGICAL EFFECTS OF THYROID HORMONES

Thyroid hormones exert effects on virtually every organ system. Their overarching action is to increase the basal metabolic rate (BMR). Complete absence of thyroid hormones causes the BMR to fall 40–50% below normal, while extreme excess can increase it 60–100% above normal.

System	Effect of Thyroid Hormones	Clinical Parallel
Metabolism (BMR)	↑ O_2 consumption, ↑ heat production, ↑ glucose absorption, ↑ gluconeogenesis, ↑ lipolysis, ↓ cholesterol (↑ LDL receptors)	Hyperthyroid: weight loss, heat intolerance; Hypothyroid: weight gain, cold intolerance, hypercholesterolaemia
Cardiovascular	↑ HR, ↑ CO, ↑ myocardial contractility, ↓ SVR, ↑ blood volume; sensitises heart to catecholamines	Hyperthyroid: palpitations, AF, ↑ pulse pressure; Hypothyroid: bradycardia, pericardial effusion
Respiratory	↑ RR, ↑ sensitivity of respiratory centre to CO_2, maintain normal hypoxic drive	Hypothyroid: hypoventilation, myxoedema coma (↓ respiratory drive)
CNS	Essential for brain development (foetal/neonatal); maintains mentation, mood, and reflexes in adults	Cretinism (congenital deficiency): irreversible intellectual disability. Adults: hypothyroid → slow cognition, depression; Hyperthyroid → anxiety, tremor

System	Effect of Thyroid Hormones	Clinical Parallel
GI	↑ Gut motility, ↑ appetite, ↑ gastric acid secretion	Hyperthyroid: diarrhoea, ↑ appetite; Hypothyroid: constipation, anorexia
Musculoskeletal	↑ Protein synthesis (physiological); excess causes protein catabolism, proximal myopathy	Hyperthyroid: myopathy, osteoporosis (↑ bone turnover); Hypothyroid: myalgia, raised CK
Reproductive	Required for normal sex hormone synthesis and metabolism; maintains libido and fertility	Hypothyroid: menorrhagia, anovulation, infertility, hyperprolactinaemia
Growth & Development	Synergises with GH; essential for bone maturation and linear growth	Hypothyroid children: short stature, delayed bone age, delayed puberty
Haemopoiesis	Stimulates EPO production; ↑ 2,3-DPG in RBC (improves O_2 release)	Hypothyroid: normocytic or macrocytic anaemia

§5 REGULATION OF THYROID HORMONE SECRETION

5.1 The Hypothalamo-Pituitary-Thyroid (HPT) Axis

Thyroid hormone secretion is controlled by a classic negative feedback loop:

- The **hypothalamus** secretes Thyrotropin-Releasing Hormone (TRH) into the portal blood.
- TRH stimulates thyrotrophs in the **anterior pituitary** to release Thyroid-Stimulating Hormone (TSH).
- TSH binds to TSH receptors on thyrocytes, activating adenylyl cyclase (Gs-coupled → cAMP → PKA). This stimulates all steps of thyroid hormone synthesis and secretion, and also promotes thyroid growth (trophic effect).
- Rising T3/T4 feed back to **suppress both TRH and TSH** via inhibitory effects on the hypothalamus and pituitary. T3 (converted from T4 within pituitary thyrotrophs by type 2 deiodinase) is the primary feedback signal.
- **Cold exposure** → hypothalamic TRH release → ↑ TSH → ↑ T3/T4 → thermogenesis.
- **Anxiety/psychological stress** → via sympathetic nervous system → ↓ TSH → ↓ T3/T4.

5.2 Autoregulation by Iodine

Iodine State	Effect on Thyroid
Iodine Deficiency	Hypertrophy (goitre); preferential T3 secretion; low T4; elevated TSH
Acute Iodine Loading (Wolff-Chaikoff effect)	Transient inhibition of organification → ↓ hormone synthesis. Normal gland escapes within 10–14 days (down-regulation of NIS)
Persistent high iodine	In susceptible individuals (e.g., Hashimoto's): may fail to escape → hypothyroidism. Or can trigger hyperthyroidism (Jod-Basedow effect) in multinodular goitre

Iodine State	Effect on Thyroid
Lugol's iodine (pre-op)	Reduces vascularity and hormone release — used for 10–14 days before thyroidectomy

5.3 Exogenous Factors Affecting Thyroid Function

Factor	Effect on TSH	Effect on T3/T4
Smoking	↓ TSH	↑ T3 and T4
Alcohol excess	↑ TSH	↓ Free T3
Obesity	↑ TSH	↑ Free T3 (mild)
High iodine intake	↑ TSH (if hypothyroidism develops)	Variable
Perchlorate (pollutant)	Normal	↓ T3/T4 (blocks NIS)
Amiodarone	Variable	Rich in iodine; inhibits T4→T3 conversion; can cause hypo- or hyper-thyroidism
Lithium	↑ TSH (if hypothyroidism)	↓ T3/T4 (blocks hormone release)

§6 MECHANISM OF ACTION

6.1 Nuclear (Genomic) Mechanism

T3 (the active form) diffuses across the plasma membrane and binds to Thyroid Hormone Receptors (TRα and TRβ), which are members of the nuclear receptor superfamily. The TR forms a heterodimer with the Retinoid X Receptor (RXR) and binds to Thyroid Hormone Response Elements (TREs) in the promoter regions of target genes.

- In the **absence of T3**: TR-RXR bound to TRE recruits co-repressors, silencing gene transcription (active repression).
- In the **presence of T3**: T3 binding causes conformational change → co-repressors dissociate → co-activators (e.g., SRC-1, GRIP-1) recruited → gene transcription initiated.
- This process is **slow** (hours to days) but produces sustained effects via new protein synthesis (e.g., Na^+/K^+-ATPase subunits, myosin heavy chains, GLUT4).

6.2 Non-Genomic Actions

T3 and T4 also exert rapid (seconds to minutes) non-genomic effects at the plasma membrane, cytoplasm, and mitochondria. These involve: activation of intracellular signalling cascades (PI3K, MAPK, PKA, PKC) and direct effects on ion channels and transporters. These non-genomic actions account for the rapid cardiovascular effects of thyroid hormones and may mediate some angiogenic effects of T4.

§7 DISORDERS OF THE THYROID GLAND

7.1 Hyperthyroidism (Thyrotoxicosis)

Hyperthyroidism results from excess circulating thyroid hormones. The clinical features reflect an exaggeration of normal thyroid hormone physiology — an acceleration of metabolism and sensitisation of target organs to catecholamines.

Feature	Graves' Disease	Toxic Multinodular Goitre (MNG)	Toxic Adenoma
Mechanism	TSI (thyroid-stimulating immunoglobulins) → autonomous TSH-receptor activation	Multiple autonomous nodules develop over time	Single autonomous nodule; suppresses rest of gland
Pathology	Diffuse goitre; autoimmune	Lumpy, nodular goitre	Solitary 'hot' nodule on scan
Specific features	Ophthalmopathy (exophthalmos/proptosis); pretibial myxoedema; thyroid acropachy	Older patients; common in iodine-deficient areas	Often single firm nodule palpable
TSH	Suppressed (<0.1 mU/L)	Suppressed	Suppressed
T3/T4	Elevated	Elevated (T3 often predominates)	Elevated
Thyroid scan (^{123}I)	Diffuse uptake	Patchy, heterogeneous uptake	Solitary hot nodule; rest cold
Treatment	Antithyroid drugs (PTU/carbimazole), radioiodine, thyroidectomy	Radioiodine preferred; surgery	Radioiodine or surgery

SYMPTOMS OF HYPERTHYROIDISM

Metabolic: Weight loss despite increased appetite, heat intolerance, excessive sweating.
Cardiovascular: Palpitations, tachycardia, atrial fibrillation, wide pulse pressure.
Neurological: Anxiety, irritability, tremor (fine, postural), insomnia, hyperreflexia.
Musculoskeletal: Proximal myopathy, osteoporosis (↑ bone turnover).
GI: Diarrhoea, ↑ stool frequency, ↑ appetite.
Reproductive: Oligomenorrhoea / amenorrhoea, decreased fertility.
Labs: High T3 and/or T4; suppressed TSH; may have positive TSI (Graves').

7.2 Hypothyroidism

Type	Cause	Key Features	Lab Pattern
Primary Hypothyroidism	Hashimoto's thyroiditis (autoimmune destruction, anti-TPO antibodies); post-radioiodine; post-thyroidectomy; iodine deficiency; lithium; amiodarone	Fatigue, weight gain, cold intolerance, constipation, dry skin, hair loss, slow reflexes (delayed relaxation), bradycardia, macroglossia, periorbital oedema, hoarse voice, depression, menorrhagia	High TSH; Low FT4; anti-TPO antibodies (Hashimoto's)
Secondary Hypothyroidism	Pituitary failure (e.g., Sheehan's syndrome, pituitary adenoma, cranial irradiation)	Same features as primary but milder; associated with deficiencies of other pituitary hormones	Low TSH; Low FT4 (key diagnostic distinction from primary)
Subclinical Hypothyroidism	Borderline primary failure	Often asymptomatic; may have mild symptoms	High TSH (typically 4–10 mU/L); Normal FT4
Congenital Hypothyroidism (Cretinism)	Thyroid agenesis/dysgenesis; defects in synthesis; maternal iodine deficiency	IF UNTREATED: irreversible intellectual disability, growth failure, deafness. Detected by neonatal screening (TSH)	High TSH; Low T4 on neonatal screen

7.3 Goitre

A goitre is any enlargement of the thyroid gland, regardless of the cause or functional status. It can be diffuse or nodular, and the patient may be euthyroid, hypothyroid, or hyperthyroid.

Type	Cause	Functional Status
Simple/Colloid goitre	Iodine deficiency (most common worldwide); puberty/pregnancy (physiological)	Usually euthyroid; may become toxic (MNG) over time
Autoimmune goitre	Graves' disease; Hashimoto's thyroiditis	Hyperthyroid (Graves') or hypothyroid (Hashimoto's)
Multinodular goitre (MNG)	Long-standing iodine deficiency; nodular hyperplasia	Euthyroid or hyperthyroid (toxic MNG)
Toxic goitre	Toxic MNG; toxic adenoma; Graves'	Hyperthyroid
Malignant goitre	Thyroid cancer (papillary, follicular, medullary, anaplastic)	Usually euthyroid; cold nodule on scan

EMERGENCY — THYROID STORM

Thyroid storm (thyrotoxic crisis) is a life-threatening exacerbation of hyperthyroidism, usually precipitated by stress (surgery, infection, trauma, labour). Features: fever >38.5°C, severe tachycardia (often AF), extreme agitation/psychosis, diarrhoea, vomiting, and high-output cardiac failure. Mortality >20% without treatment.
Treatment: PTU (blocks synthesis AND peripheral conversion) + Lugol's iodine (after PTU) + beta-blockers (propranolol) + hydrocortisone (prevents adrenal crisis) + supportive care.